Assignment – 3

It can be solved by using NumPy's `delete()` and `insert()` functions.

```
import numpy as np

# Original Array
arr = np.array([[1, 2, 3],
                [4, 5, 6],
                [7, 8, 9]])

# Delete column 2 (index 1)
arr1 = np.delete(arr, 1, axis=1)

print("Printing Array after deleting col 2 on axis 1")
print(arr1)

# New column to insert
new_col = np.array([0, 0, 0])

# Insert the new column at column 2 (index 1)
arr2 = np.insert(arr1, 1, new_col, axis=1)

print("Printing Array after inserting col 2 on axis 1")
print(arr2)
```

Output

```
Printing Array after deleting col 2 on axis 1
[[1 3]
 [4 6]
 [7 9]]

Printing Array after inserting col 2 on axis 1
[[1 0 3]
 [4 0 6]
 [7 0 9]]
```

- `np.delete(arr, 1, axis=1)` removes the **second column** (index `1`).
- `np.insert(arr1, 1, new_col, axis=1)` inserts a new column of zeros at **column index 1** (the second column), producing the required output.